Wigner’s semicircle law

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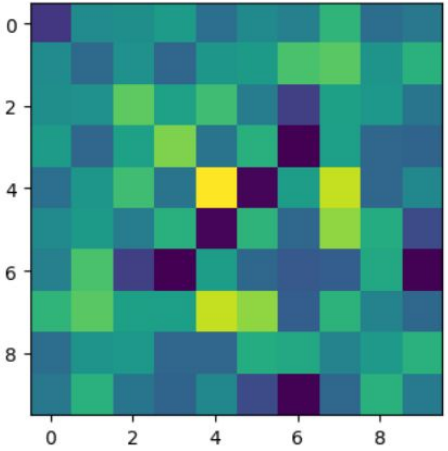
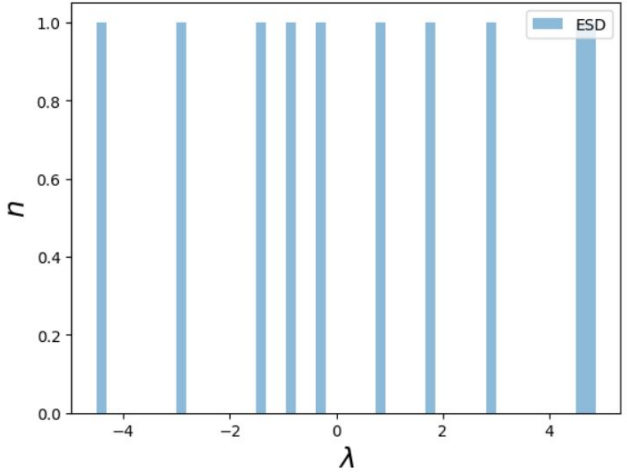
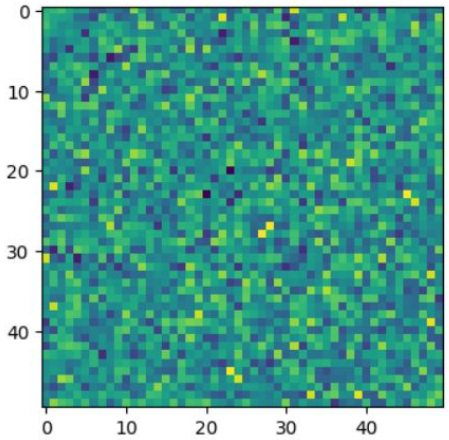
Intro

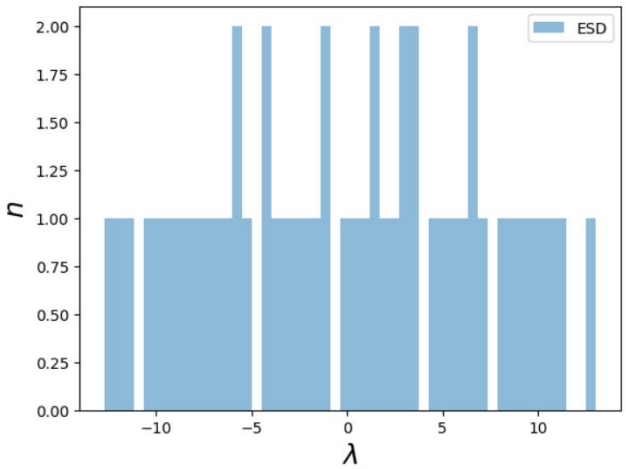




Symmetrical random matrix $\mathbf{X} \in \mathbb{R}^{n \times n}$, n = 10

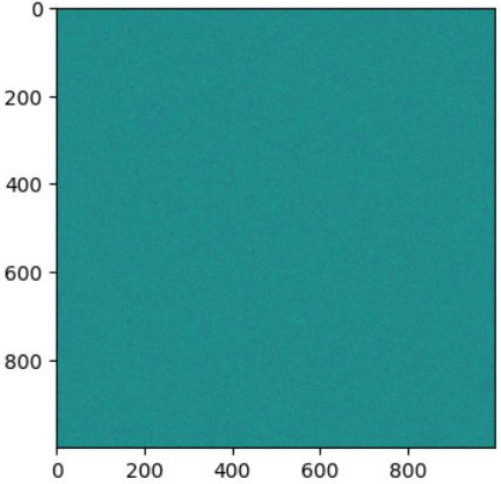
Intro

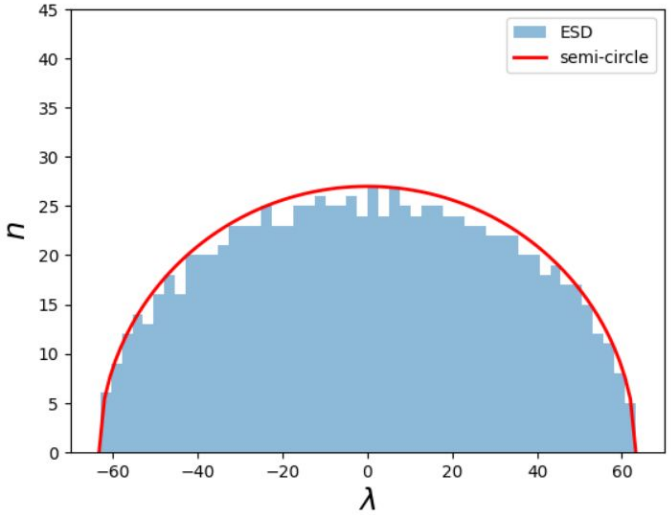




Symmetrical random matrix $\mathbf{X} \in \mathbb{R}^{n \times n}$, n = 50

Intro





Symmetrical random matrix $\mathbf{X} \in \mathbb{R}^{n \times n}$, n = 1000

Wigner matrix

Consider an infinite array of complex random variables (X_{ij}) where for $1 \leq i < j$

$$X_{ij} = \bar{X}_{ji}$$

are iid with law P on $\mathbb{C}$, independent of $X_{ii}, i \geq 1$ iid with common law Q on $\mathbb{R}$.

The random matrix $X = (X_{ij})_{1 \leq i,j \leq n}$ is hermitian. This matrix is called a *Wigner matrix*.

Important cases

- For $i > j$, $\sqrt{2}\Re(X_{ij})$, $\sqrt{2}\Im(X_{ij})$ and X_{ii} are independent standard Gaussian variables: *Gaussian Unitary Ensemble* (GUE).

- For $i > j$, X_{ij} and $X_{ii}/\sqrt{2}$ are independent standard Gaussian variables): *Gaussian Orthogonal Ensemble* (GOE).

- For $i > j$, X_{ij} and X_{ii} are independent $\{0, 1\}$-Bernoulli distribution with parameters $0 \leq p \leq 1$ and $0 \leq q \leq 1$.

Empirical distribution of eigenvalues

Let X be an hermitian matrix in $M_n(\mathbb{C})$. Counting multiplicities, its ordered eigenvalues are denoted as

$$\lambda_1(X) \geq \cdots \geq \lambda_n(X).$$

The *empirical spectral distribution (ESD)* is

$$\mu_X = \frac{1}{n}\sum_{i=1}^{n} \delta_{\lambda_i(X)}.$$

This is a global function of the spectrum. From the spectral theorem, for any function f,

$$\int f(\lambda) d\mu_X(\lambda) = \frac{1}{n}\mathrm{Tr} f(X).$$

Weak Convergence

Let $\mu_n, \mu \in \mathcal{P}(\mathbb{R})$, we say that μ_n converges weakly to μ if for any bounded continuous function f,

$$\int f d\mu_n \to \int f\mu$$

This weak topology is metrizable with the Lévy distance d_L and $(\mathcal{P}(\mathbb{R}), d_L)$ is a Polish space.

If μ_n is a random measure, we say that μ_n converges weakly in probability to μ if for any bounded continuous function f, in probability.

$$\int f d\mu_n \to \int f\mu$$

Similarly, we say that μ_n converges weakly to μ almost surely if almost surely, for any bounded continuous function f,

$$\int f d\mu_n \to \int f\mu.$$

Wigner's semicircle Theorem

In our setting, a general form of Wigner's semicircle law is the following result.

Theorem 1.1 (Semicircle Law)**.** *Let X be a Wigner matrix. Assume that $\mathbb{E}|X_{12} - \mathbb{E}X_{12}|^2 = 1$. Set $Y = X/\sqrt{n}$. Then a.s. weakly*

$$\mu_Y \to \mu_{sc}$$

where $\mu_{sc}(dx) = \frac{1}{2\pi}\sqrt{4 - x^2}\mathbf{1}_{|x|\leq 2}dx$

Method of moments

Let Z real random variable with law μ and characteristic function $F(t) = \int e^{itx} d\mu(x)$. Assume that all moments are finite: for all integers $k \geq 0$, $\int x^k d\mu(x) = \mathbb{E}Z^k = m_k < \infty$.

Definition 1.1. *The probability measure μ is uniquely characterized by its moments if it is the unique measure on $\mathbb{R}$ with moments m_k for all integers $k \geq 0$.*

From Carleman's Theorem, μ is uniquely characterized by its moments if

$$\sum_{k \geq 1} m_{2k}^{-\frac{1}{2k}} = \infty.$$

Method of moments

Lemma 1.1. *The characteristic function F is analytic in a complex neighborhood of* 0 *(or in a complex neighborhood of* $\mathbb{R}$*) if and only if*

$$\limsup_k \left(\frac{m_{2k}}{(2k)!} \right)^{\frac{1}{2k}} < \infty.$$

Proof

Proof. Hadamard's formula: the radius of convergence of $\sum_n a_n z^n$ associated to a sequence of complex numbers (a_n) is $1/r$ with $r = \limsup_n |a_n|^{1/n}$. It gives the equivalence. To show analiticity on $\mathbb{R}$, we may use Taylor expansion

$$F(t+s) = \sum_{\ell=0}^{2k-1} \frac{(it)^\ell F^{(\ell)}(s)}{\ell!} + O\left(\frac{m_{2k}|t|^{2k}}{(2k)!}\right).$$

Stirling's formula:

$$k! \sim \sqrt{2\pi k}(k/e)^k$$

Example

Stirling's formula:

$$k! \sim \sqrt{2\pi k}(k/e)^k$$

- if μ has bounded support : $\mu_k \leq c^k$
- If μ is a subGaussian law: $\mu_k \leq (c\sqrt{k})^k$
- If μ is subexponential: $\mu_k \leq (ck)^k$

In these cases, μ is uniquely characterized by its moments.

for all integer $k \geq 1$, $\lim_n \mathbb{E}[Z_n^k] = \mathbb{E}[Z^k] \implies (Z_n)_{n\geq 1}$ converges weakly to Z

Method of moments

Lemma 1.2. *Assume that the law μ is uniquely determined by its moments.*

If for all integers $k \geq 1$, $\lim_n \int x^k d\mu_n(x) = \int x^k d\mu$ then $\mu_n \to \mu$ weakly.

Proof. We have $\int x^2 d\mu(x) = m_2 + o(1)$, hence (μ_n) is relatively compact. Let ν be an accumulation point, since μ is uniquely determined by its moments, it is sufficient to check that for any integer $k \geq 1$, $\int x^k d\nu(x) = m_k$. This amounts to prove that $x \mapsto x^k$ is uniformly integrable for $(\mu_n)_{n\geq 1}$.

Moments of Wigner's semicircle law

$$d\mu_{sc}(x) = \frac{\mathbf{1}_{|x|\leq 2}}{2\pi}\sqrt{4-x^2}dx$$

For integer $k \geq 0$,

$$m_{2k+1} = \int x^{2k+1} d\mu_{sc}(x) = 0 \quad \text{and} \quad \boxed{m_{2k} = \int x^{2k} d\mu_{sc}(x) = c_k}$$

where c_k is the k-th Catalan number

$$c_k = \frac{1}{k+1}\binom{2k}{k} = \binom{2k}{k} - \binom{2k}{k+1}.$$

Graphs, Catalan's number and Dyck paths

Fix an integer $k \geq 1$. We set of *Dyck paths* of length $2k$ is the set of paths of length $2k$ from 0 to 0 with ± 1 increments and which remain non-negative:

$$\mathcal{D}_k = \{(x_0, \ldots, x_{2k}) \in \mathbb{Z}^{2k+1} : x_t \geq 0, x_0 = x_{2k+1}, x_{t+1} - x_t \in \{-1, 1\} \text{ for } t = 1, \ldots, 2k\}.$$

$\mathcal{D}_k$ is an obvious bijection with sequence of proper sequence of $2k$ parenthesis: for $t = 1, \ldots, 2k$, if $x_t = x_{t-1} + 1$ place a left parenthesis '(' and right parenthesis ')' otherwise. For example, for $k = 3$, the path $(0, 1, 0, 1, 2, 1, 0)$ gives $()(())$.

Dyck's paths

Lemma 1.4. *For any integer $k \geq 1$,*

$$|\mathcal{D}_k| = c_k.$$

Proof. Let $\mathcal{W}_k^s$ be set of walks of length $2k$ from 0 to even integer $-2s$ with increments $\{-1, +1\}$, $\mathcal{W}_k^s = \{(x_0, \ldots, x_{2k}) \in \mathbb{Z}^{2k+1} : x_0 = 0, x_{2k+1} = -2s, x_{t+1} - x_t \in \{-1, 1\} \text{ for } t = 0, \ldots, 2k\}$. We have $|\mathcal{W}_k^s| = \binom{2k}{k+s}$ since such walks are made of $k - s$ times t such that $x_{t+1} - x_t = 1$ and $k + s$ times such that $x_{t+1} - x_t = -1$. We check that $|\mathcal{D}_k| = |\mathcal{W}_k^0| - |\mathcal{W}_k^1|$. This is a consequence of the reflexion principle: paths in $\mathcal{W}_k^1$ are in bijections with paths in $\mathcal{W}_k$ such that $x_t = -1$ for some t (simply reflects x_{t+s} for $s \geq 0$ on x-axis). $\square$

Quick recap: spectral theorem

(?) $\mathbb{E}\mu_{A_N} \to \mu_{sc}$

The *empirical spectral distribution (ESD)* is

$$\mu_X = \frac{1}{n}\sum_{i=1}^{n} \delta_{\lambda_i(X)}.$$

This is a global function of the spectrum. From the spectral theorem, for any function f,

$$\int f(\lambda) d\mu_X(\lambda) = \frac{1}{n}\mathrm{Tr} f(X).$$

Recall that, for a random probability measure $\mu \in \mathcal{P}(\mathcal{X})$, we may define its expectation $\mathbb{E}\mu \in \mathcal{P}(\mathcal{X})$, as for all Borel sets B,

$$(\mathbb{E}\mu)(B) = \mathbb{E}[\mu(B)].$$

Thus, for a Wigner's matrix defined as $A_N = \frac{1}{\sqrt{N}}(a_{ij})_{i,j=1}^{N}$, using a spectral theorem we have:

$$\boxed{\mathbb{E}\int \lambda^m d\mu_{A_N}(\lambda) = \frac{1}{N}\mathbb{E}TrA_N^m = \boxed{\mathbb{E}[tr(A_N^m)]}} \tag{1}$$

Wigner's semicircle law for averaged eigenvalues

Consider a Wigner matrix $A_N = \frac{1}{\sqrt{N}} (a_{ij})_{i,j=1}^N$, where μ has all moments and

$$\int_{\mathbb{R}} x \, \mathrm{d}\mu(x) = 0, \qquad \int_{\mathbb{R}} x^2 \, \mathrm{d}\mu(x) = 1.$$

Theorem 3.9 (Wigner's semicircle law for Wigner matrices, averaged version)**.** *Let A_N be a Wigner matrix corresponding to μ, which has all moments, with mean 0 and second moment 1. Then we have for all $m \in \mathbb{N}$:*

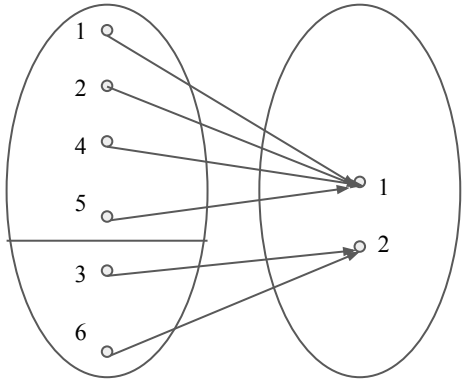
$$\lim_{N\to\infty} \boxed{\mathbb{E}\left[\operatorname{tr}\left(A_N^m\right)\right]} = \frac{1}{2\pi} \int_{-2}^{2} x^m \sqrt{4 - x^2} \, \mathrm{d}x.$$

Source: Prof. Dr. Roland Speicher - Lecture notes on Random matrices, Saarland University

Proof

$$\boxed{\mathbb{E}\left[\operatorname{tr}(A_N^m)\right]} = \frac{1}{N^{1+\frac{m}{2}}} \sum_{i_1,\ldots,i_m=1}^{N} \mathbb{E}\left[a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_m i_1}\right]$$

Motivation: grouping the multi-indices $i = (i_1, \ldots, i_m)$ by their kernels



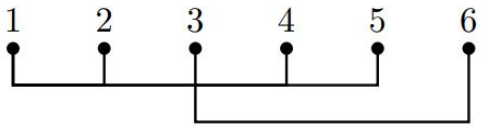
For a multi-index $i = (i_1, \ldots, i_m)$ we denote by $\ker i$ its *kernel*; this is the partition $\sigma \in \mathcal{P}(m)$ such that we have $i_k = i_l$ if and only if k and l are in the same block of σ. If we identify i with a function $i\colon [m] \to [N]$ via $i(k) = i_k$ then we can also write

$$\ker i = \left\{ i^{-1}(1), i^{-1}(2), \ldots, i^{-1}(N) \right\},$$

where we discard all empty sets.

Example

For example, for $i = (1, 1, 2, 1, 1, 2)$ and $j = (2, 2, 7, 2, 2, 7)$ we have

$\ker i =$ 

$= \ker j$

$$\mathbb{E}\left[a_{11}a_{12}a_{21}a_{11}a_{12}a_{21}\right] = \mathbb{E}\left[a_{11}^2\right]\mathbb{E}\left[a_{12}^4\right] = \mathbb{E}\left[a_{22}^2\right]\mathbb{E}\left[a_{27}^4\right] = \mathbb{E}\left[a_{22}a_{27}a_{72}a_{22}a_{27}a_{72}\right]$$

We denote this common value by

$$\mathbb{E}\left[\sigma\right] := \mathbb{E}\left[a_{i_1 i_2}a_{i_2 i_3}\cdots a_{i_m i_1}\right] \qquad \text{if } \ker i = \sigma.$$

Proof

Then

$$\boxed{\mathbb{E}\left[\operatorname{tr}(A_N^m)\right]} = \frac{1}{N^{1+\frac{m}{2}}} \sum_{i_1,\ldots,i_m=1}^{N} \mathbb{E}\left[a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_m i_1}\right] = \frac{1}{N^{1+\frac{m}{2}}} \sum_{\sigma \in \mathcal{P}(m)} \boxed{\sum_{\substack{i\colon [m]\to[N] \\ \ker i = \sigma}} \mathbb{E}\left[\sigma\right]},$$

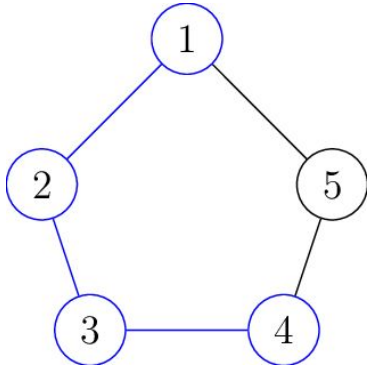
$$\boxed{\mathbb{E}\left[\operatorname{tr}(A_N^m)\right]} = \frac{1}{N^{1+\frac{m}{2}}} \sum_{\sigma \in \mathcal{P}(m)} \mathbb{E}\left[\sigma\right] \cdot \boxed{\#\left\{i\colon [m] \to [N] \mid \ker i = \sigma\right\}}.$$

$$\#\left\{i\colon [m] \to [N] \mid \ker i = \sigma\right\} = N(N-1)(N-2)\cdots(N-\#\sigma+1),$$

In order to understand the contribution corresponding to each of the partitions, we will introduce a graph as follows

Proof

For $\sigma = \{V_1, \ldots, V_k\} \in \mathcal{P}(m)$ we define a corresponding graph $\mathcal{G}_\sigma$ as follows. The vertices of $\mathcal{G}_\sigma$ are given by the blocks V_p of σ and there is an edge betwees V_p and V_q if there is an $r \in [m]$ such that $r \in V_p$ and $r + 1 (\mathrm{mod}\ m) \in V_q$.



The term $\mathbb{E}\left[a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_m i_1}\right]$ corresponds now to a walk in $\mathcal{G}_\sigma$, with $\sigma = \ker i$, along the edges with steps

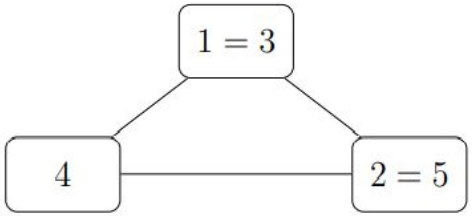
$$i_1 \to i_2 \to i_3 \to \cdots \to i_m \to i_1.$$

$\sigma = \{(1,3), (2,5), (4)\} =$

Only such walks/terms that remain in the summation are the ones where each edge is visited twice, and thus:

$$\#\text{ edges in } \mathcal{G}_\sigma \leq \frac{\#\text{ steps in the walk}}{2} = \frac{m}{2}.$$

$\mathcal{G}_\sigma =$



$$\boxed{\mathbb{E}\left[\mathrm{tr}\left(A_N^m\right)\right]} = \frac{1}{N^{1+\frac{m}{2}}} \sum_{\substack{\sigma \in \mathcal{P}(m) \\ \#\mathrm{edges}(\mathcal{G}_\sigma) \leq \frac{m}{2}}} \mathbb{E}\left[\sigma\right] \underbrace{N(N-1)(N-2)\cdots(N-\#\sigma+1)}_{\sim N^{\#\sigma} \text{ for } N \to \infty}. \qquad (\star)$$

Proof

Proposition 3.8. *Let $\mathcal{G} = (V, E)$ be a connected finite graph with vertices V and edges E. (We allow loops and multi-edges.) Then we have that*

$$\#V \leq \#E + 1$$

and we have equality if and only if $\mathcal{G}$ is a tree, *i.e., a connected graph without cycles.*

$$\lim_{N\to\infty} \boxed{\mathbb{E}\left[\operatorname{tr}\left(A_N^m\right)\right]} = \sum_{\sigma\in\mathcal{P}(m)} \mathbb{E}\left[\sigma\right] \lim_{N\to\infty} N^{\#V(\mathcal{G}_\sigma)-\frac{m}{2}-1}.$$

$$\#V(\mathcal{G}_\sigma) \leq \#E(\mathcal{G}_\sigma) + 1 \leq \frac{m}{2} + 1. \quad \xrightarrow{N\to\infty} \quad \boxed{\#V(\mathcal{G}_\sigma) = \#E(\mathcal{G}_\sigma) + 1 = \frac{m}{2} + 1.}$$

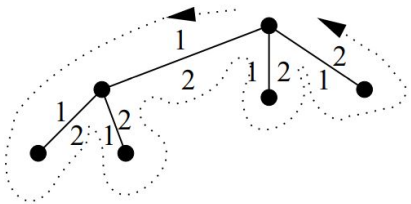
Thus, $\mathcal{G}_\sigma$ must be a tree and in our walk we use each edge exactly twice $\mathbb{E}\left[\sigma\right] = 1$

$$\lim_{N\to\infty} \boxed{\mathbb{E}\left[\operatorname{tr}\left(A_N^m\right)\right]} = \#\left\{\sigma \in \mathcal{P}(m) \,|\, \mathcal{G}_\sigma \text{ is a tree}\right\}$$

Proof

Number of partitions σ ∈ P(m), for which the corresponding graph Gσ is a tree is also counted by the Catalan numbers!

$w = 1121221212$



- Given Dyck word w, form an ordered tree as follows:
  - Draw the root.
  - Read w from left to right.
    For 1, add a new rightmost child to the current vertex and move to it.
    For 2, go up to the parent of the current vertex.
- For any prefix of w with a 1's and b 2's, the depth of the vertex you reach is $a - b \geqslant 0$, so you do not go "above" the root.
  At the end, $a = b = n$ and the depth is $a - b = 0$ (the root).
- Conversely, trace an ordered tree counterclockwise from the root. Label each edge 1 going down its left side, and 2 going up its right.
- Thus, W_n is in bijection with ordered trees on n edges (hence $n + 1$ vertices), so C_n counts these too.

Thus: $$\lim_{N\to\infty} \mathbb{E}\left[\operatorname{tr}\left(A_N^m\right)\right] = \frac{1}{2\pi}\int_{-2}^{2} x^m\sqrt{4-x^2}\,\mathrm{d}x.$$

$$\mathbb{E}\mu_{A_N} \to \mu_{sc}$$ □

Source: Topics in trees and Catalan numbers, Prof. Tesler, UCSD